University College London, 2025-2026
Econ 0016 – International Macroeconomics – Franck Portier

Problem set 3 : Uncertainty and the Current Account – The Twin Deficits: Fiscal Deficits and the Current Account

Chapter 6 and 8 in Schmitt-Grohé, Uribe and Woodford

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True/False/Uncertain

1. The risk-free interest rate goes up with uncertainty in period 2 endowment Q_2 in an small open economy.

2. All else constant, the more complete international financial markets are, the bigger the gap between output and consumption volatility will be.

3. A current tax cut with current and future government spending unchanged causes an increase in saving as opposed to an increase in consumption because households understand that taxes will have to increase in the future. Since the current account equals saving minus investment, the current account improves.

Exercise 6.4 (The Current Account as Insurance against Catastrophic Events)

Consider a two-period endowment economy populated by identical households with preferences defined over consumption in period 1, C_1, and consumption in period 2, C_2, and described by the utility function

$$\log C_1 + E[\log C_2],$$

where E denotes the expected value operator. Each period, households receive an endowment of 10 units of food. Households start period 1 carrying no assets or debts from the past ($B_0 = 0$). Financial markets are incomplete. There is a single internationally traded bond that pays the interest rate $r^\star = 0$

1. Compute consumption, the trade balance, the current account, and national saving in period 1.

2. Assume now that the endowment in period 1 continues to be 10, but that the economy is prone to severe natural disasters in period 2. Suppose that these negative events are very rare, but have catastrophic effects on the country's output. Specifically, assume that with probability 0.01 the economy suffers an earthquake in period 2 that causes the endowment to drop by 90 percent with respect to period 1. With probability 0.99, the endowment in period 2 is 111/11. What is the expected endowment in period 2? How does it compare to that of period 1?

3. What percent of period 1 endowment will the country export? Compare this answer to what happens under certainty and provide intuition.

4. Suppose that the probability of the catastrophic event increases to 0.02, all other things equal. Compute the mean and standard deviation of the endowment in period 2. Is the change in probability mean preserving?

5. Calculate the equilibrium levels of consumption and the trade balance in period 1.

6. Compare your results with those pertaining to the case of 0.01 probability for the catastrophic event. Provide interpretation

Exercise 8.6 (Imperfect Capital Mobility and Crowding Out)

Consider a small open economy with identical households whose preferences are described by the utility function

$$\log C_1 + \beta \log C_2,$$

where C_1, C_2 ,and $\beta = 0.96$ denote consumption in period 1, consumption in period 2, and the subjective discount factor. Households have an endowment of $Q1 = 20$ units of goods in period 1 and receive profits Π_2 from firms in period 2. They enter period 1 with no assets or debts ($B_0^h = 0$), and can borrow or lend at the interest rate r_1. In period 1, firms borrow at the rate r_1 to invest in I_1 units of capital goods that become productive in period 2. In period 2, they use the production technology

$$6\sqrt{I_1}$$

to produce goods, they pay back their loans, and distribute profits to households. The government enters period 1 with zero debts or assets ($B_0^g = 0$), spends $G_1 = 1$ in period 1, and $G_2 = 7$ in period 2 on goods and levies lump-sum taxes $T_1 = G_1$ and $T_2 = G_2$. The interest rate at which the rest of the world is willing to lend to this country is

$$r_1 = \begin{cases} r^\star & \text{if } B_1 \geq 0 \\ r^\star + p & \text{if } B_1 < 0, \end{cases}$$

where $r^\star$ is the world interest rate paid to net external creditors and equals 8 percent($r^\star = 0.08$), p is an interest rate premium charged to net external debtors and equals 2 percent($p = 0.02$), and B_1 is the country's net foreign asset position at the end of period 1

1. Calculate the equilibrium values of the interest rate r_1, the current account in period 1, CA_1, and investment in period 1, I_1.

2. Suppose now that because of extraordinary expenses stemming from the COVID-19 pandemic, government spending in period 1 increases by 100 percent (i.e., G_1 goes from 1 to 2). Recalculate r_1, CA_1, and I_1. Does government spending crowd out investment?

3. Suppose now that the increase in government spending in period 1 is not 100 percent but 300 percent (i.e., G_1 goes from 1 to 4). Recalculate r_1, CA_1, and I_1. Does government spending crowd out investment? Provide intuition.

4. Continue to assume that $G_1 = 4$, but suppose that the country premium is 4 percent($p = 0.04$). Show numerically that an equilibrium does not exist. Explain your finding using a graph